Supraja Srikanth

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SUMMARY

Software Engineering **Master's** candidate with **2+** years of experience building scalable distributed systems and AI-driven products at **Amazon** and **Thoughtworks**. Expert in architecting high-throughput backend infrastructures and ML pipelines using **Python**, **Java**, and **PyTorch**. Proven track record of optimizing system latency by 75ms and leading cross-functional technical efforts to ship production-grade RAG systems and microservices

EDUCATION

University of Cincinnati | Cincinnati, OH **Dec 2026**

Master of Science in Computer Science, **GPA: 3.6**

Relevant Courses: **Artificial Intelligence (AI)**, **Machine Learning**, Data Structures and Algorithms

VNR VJIET | Hyderabad, India **Jun 2023**

Relevant Courses: Software Engineering, Cloud Computing, Python, Machine Learning

EXPERIENCE

Thoughtworks– Application Developer | Hyderabad, India **Jan 2024 – Jul 2025**

- Engineered bill notifications and alerts within the distributed autopay platform using **Spring Boot and Java**, leveraging **software design patterns** to build reliable, modular components, and boosting **customer experience by 30%** through consistent payment reminders.
- Architected a robust recurring mandate protocol for autopay, confirming instantaneous payments after bill production, thereby maximizing payment efficiency and **diminishing payment failure rates by 1.5%.**
- Decreased payment processing latency by **75ms** by optimizing BBPS APIs; accelerated bill and biller data retrieval, enhancing performance, scalability, and customer satisfaction in real-time transactions.
- Implemented **Redis Pub/Sub architecture**, increasing concurrent processing capabilities by at least **25%**, decreasing execution latency for **high-demand distributed workflows**, and improving overall system responsiveness.
- Partnered with product managers and designers to ship user-facing payment experiences, iterating quickly based on customer needs and production feedback.

KAS Commerce - Software Engineer | Mumbai, India **Oct 2023 - Jan 2024**

- Optimized API response time for Stripe transactions by **150ms**, tracking **key performance indicators (KPIs)** on transaction latency to enhance user experience during checkout.
- Implemented Amazon OAuth for user registration and authentication, reducing manual verification time by **40%**, accelerating onboarding, and improving system performance, security, and user experience.
- Built interactive **business intelligence** dashboards using React and PostgreSQL for **data visualization**, facilitating faster data-driven decisions and increasing dashboard usage by **30%**.

Amazon - Software Engineer Intern | Hyderabad, India **Jun 2023 - Jul 2024**

- Designed a shadow testing environment with Java, Spring Boot, and AWS, augmenting code coverage by 8%; also refactored legacy code, enhancing maintainability and reducing **technical debt by 12%.**
- Debugged rate card logic leveraging JUnit, Mockito, and EasyMock for comprehensive integration testing; **elevated test coverage** by **15%**, resulting in a substantial reduction of post-deployment bugs.
- Initiated and owned an automated approval workflow in CI/CD pipelines to proactively detect regional compliance issues, improving deployment safety at scale.

TECHNICAL SKILLS

Machine Learning:	**Deep learning**, **Tensorflow**, **PyTorch**, Embeddings, and Semantic Search
Languages:	**Python libraries: NumPy, Pandas, Matplotlib**, Java , C++ , JavaScript , TypeScript , and SQL
Backend & Systems:	Distributed Systems, Microservices , Spring Boot , FastAPI , Node.js , and Redis (Pub/Sub).
Databases:	PostgreSQL, MySQL, DynamoDB, MongoDB
Cloud & DevOps:	**GCP**. AWS (EC2, S3, Lambda, RDS, IAM) ,Terraform , GitHub Actions , and CI/CD pipelines
Frontend:	React, HTML and CSS
Testing & Quality:	JUnit, Mockito, Test-Driven Development (TDD), and Integration
Other:	Performance Engineering: Runtime Analysis, Parallel Computation, Latency Optimization

PROJECTS

GPT-From-Scratch – **Transformer**, Python, PyTorch, NumPy

• Implemented Transformer-based GPT model using Python & PyTorch covering attention, & autoregressive generation.

• Built a complete training and inference pipeline, including tokenization, loss optimization, and text generation.

• Demonstrated strong understanding of LLM internals, performance considerations, and scalable model workflows.

All RAG Techniques - Python, Vector DBs, LangChain, Embeddings

• Designed RAG systems similar to **ChatGPT-style question answering**, balancing response quality, latency, and reliability.

• Implemented document ingestion, embeddings, semantic search, context injection for knowledge grounded responses**.**

• Evaluated and optimized RAG quality using retrieval accuracy, latency, and reliability metrics**.**